Practice Problems - 3

The left end of a long glass rod of index 1.6350 is ground and polished to a convex spherical surface of radius 2.50 cm. A small object is located in the air and on the axis 9.0 cm from the vertex. Find (*a*) the primary and secondary focal lengths, (*b*) the power of the surface, (*c*) the image distance, and (*d*) the lateral magnification.

Solve Prob. 3.1 graphically. (*a*) Find the image distance by the oblique-ray method 1. (*b*) Find the relative size of the image by the parallel-ray method.

The left end of a water trough has a transparent surface of radius -2.0 cm. A small object 2.5 cm high is located in the air and on the axis 10.0 cm from the vertex. Find (*a*) the primary and secondary focal lengths, (*b*) the power of the surface, (*c*) the image distance, and (*d*) the size of the image. Assume water to have an index 1.3330.

A hollow glass cell is made of thin glass in the form of an equiconcave lens. The radii of the two surfaces are 1.650 cm, and the distance between the two vertices is 1.850 cm. When sealed airtight, this cell is submerged in water of index 1.3330. Calculate (*a*) the focal lengths of each surface and (*b*) the power of each surface.